When Data Matters More: A Data-Centric Approach to Pneumonia Segmentation in Chest X-Rays

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Introduction

Pneumonia detection from chest X-rays is crucial for early diagnosis. However, automated segmentation is challenging due to noisy annotations and limited labeled data.

Problem Statement

Most segmentation studies focus on model architecture, while data quality and supervision strategies remain underexplored.

Related Work

Existing segmentation models emphasize architectural improvements while largely overlooking data quality, annotation noise, and supervision reliability.

Approach	Key Idea	Limitation
U-Net / FCN	Encoder–decoder segmentation networks	Sensitive to noisy labels
Attention U-Net	Attention modules for lesion localization	Requires large labeled datasets
SSL-based learning	Uses unlabeled data for representation learning	Rarely applied to segmentation
OHEM / Data Augmentation	Focus on difficult samples and data diversity	Ignores annotation reliability

Aim

Develop a data-centric framework to improve pneumonia segmentation robustness.

Objectives

- Improve learning from limited labeled data
- Reduce effect of noisy annotations
- Enhance segmentation reliability

Novelty

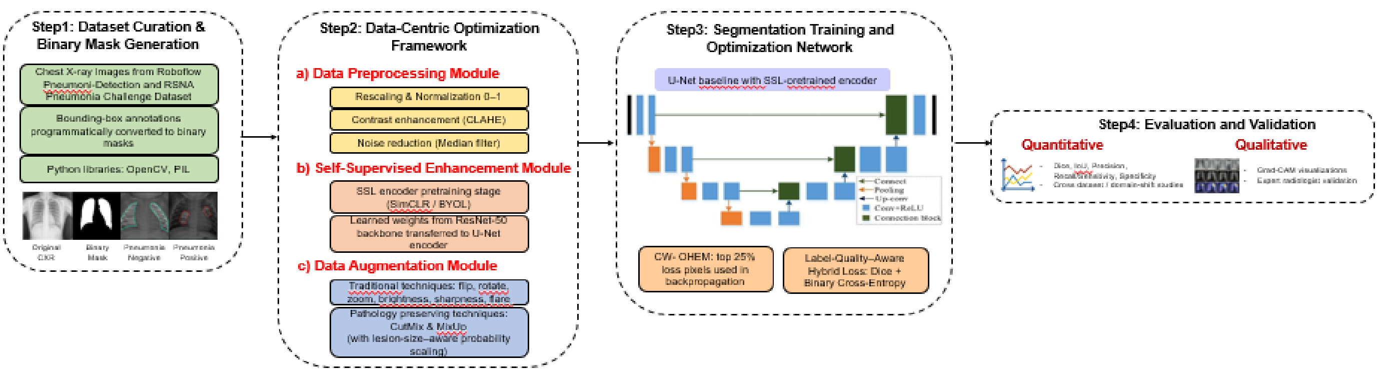
This work introduces a data-centric segmentation framework combining:

- Self-supervised representation learning
- Pathology-aware data augmentation
- Confidence-weighted hard example training

Conclusion

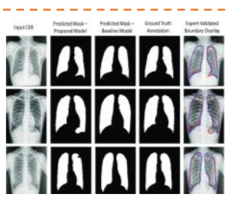
Data-centric optimization significantly improves pneumonia segmentation performance and robustness without increasing architectural complexity.

Methodology



Results and Discussion

Pneumonia detection from chest X-rays is crucial for early diagnosis. However, automated segmentation is challenging due to noisy annotations and limited labeled data.
Predicted segmentation masks accurately localize pneumonia regions, while Grad-CAM visualizations confirm clinically relevant model attention.



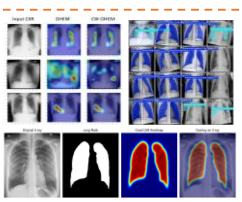


Table1: Comparison of data-centric vs architectural frameworks

Model	Setting	Dice (%)	IoU (%)	Precision (%)	Recall / Sensitivity (%)	Specificity (%)
DeepLabv3+	Model-Centric	84.9	78.4	86.1	83.5	88.0
	Data-Centric	90.3	86.3	91.8	89.1	91.5
Attention U-Net	Model-Centric	86.2	79.7	87.5	85.3	88.5
	Data-Centric	90.4	86.5	91.9	89.2	91.7
Swin-UNet	Model-Centric	87.4	81.2	88.7	86.5	89.1
	Data-Centric	90.6	86.8	92.0	89.4	91.9
U-Net	Model-Centric	83.6	77.1	85.3	82.2	87.6
	Data-Centric	90.2	86.7	91.7	89.5	91.8